

Отчёт по лабораторной работе

Лабораторная работа №3

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1 Цель работы

Целью работы является практика создания литературного стиля кода на примере модели daisyworld.

2 Задание

Выполнить работу и получить навыки создания литературного стиля.

3 Теоретическое введение

Модель Daisyworld, предложенная Джеймсом Лавлоком и Эндрю Уотсоном, иллюстрирует гипотезу Геи [2,3]. Гипотеза Геи рассматривает планету как единую, саморегулирующуюся систему, включающую как живые, так и неживые части.

В мире маргариток произрастают чёрные и белые маргаритки. Их альbedo различается: чёрные маргаритки поглощают свет и тепло, нагревая окружающую среду; белые маргаритки делают наоборот. Маргаритки могут размножаться только в определённом температурном диапазоне, а это значит, что слишком много (или слишком мало) тепла от солнца и/или окружающей среды в конечном итоге остановит их размножение.

Поверхность Геи нагревается солнцем, но растущие на ней маргаритки поглощают или отражают свет звёзд, изменяя локальную температуру.

Если температура становится слишком высокой или слишком низкой, маргаритки не захотят размножаться. Пока температура благоприятна, маргаритки конкурируют за землю и пытаются дать начало новому растению в местах, расположенных неподалёку от них.

Когда климат слишком холодный, черным маргариткам необходимо размножаться, чтобы повысить температуру, и наоборот — когда климат слишком тёплый, необходимо производить больше белых маргариток, чтобы охладить климат.

Взаимодействие живых и неживых аспектов этого мира находит равновесие в широком диапазоне параметров, хотя при достаточно сильном внешнем воздействии маргаритки не смогут регулировать температуру планеты и в

конечном итоге вымрут. Описание модели Daisyworld

В модели Daisyworld агентами являются чёрные и белые маргаритки. Они живут на клеточной сетке (среда).

Устанавливаем все необходимые пакеты (рис. 4.3) (рис. 4.4).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$ cd ../
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject$ julia --project=. -e 'using Pkg; Pkg.add(["Agents", "DataFrames", "Plots"])'
Updating registry at '~/.julia/registries/General.toml'
Resolving package versions...
Installed Statistics — v1.11.5
Installed SimpleTraits — v0.9.6
Installed OnlineStatsBase — v1.7.3
Installed StructArrays — v0.7.3
Installed StreamSampling — v0.8.2
Installed Agents — v7.0.3
```

Рисунок 4.3: Установка:

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject$ julia --project=. -e 'using Pkg; Pkg.add("StatsBase")'
Resolving package versions...
Updating ~/2026-1--study--simulation-modeling/labs/lab03/project/Project.toml
[2913bbd2] + StatsBase v0.34.13
Manifest No packages added to or removed from '~/2026-1--study--simulation-modeling/lab
s/lab03/project/Manifest.toml'
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject$
```

Рисунок 4.4: Установка:

ДСоздаём и выполняем scripts/daisyworld.jl (рис. 4.5).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject$ julia --project=. scripts/daisyworld.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject$
```

Рисунок 4.5: Выполнение:

Создаём литературный стиль (чистый код, markdown, notebook) (рис. 4.6) (рис. 4.7) (рис. 4.8).

```
julia> using Literate
julia> Literate.markdown("scripts/daisyworld.jl", "docs")
[ Info: generating markdown page from '~/2026-1--study--simulation-modeling/labs/lab03/proj
ect/scripts/daisyworld.jl'
[ Info: writing result to '~/2026-1--study--simulation-modeling/labs/lab03/project/docs/dai
syworld.md'
~/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.md"
```

Рисунок 4.6: Markdown:

```
julia> Literate.notebook("scripts/daisyworld.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/s
cripts/daisyworld.jl`
[ Info: executing notebook `daisyworld.ipynb`

[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/docs/dai
syworld.ipynb`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.ipyn
b"
```

Рисунок 4.7: Notebook:

```
julia> Literate.script("scripts/daisyworld.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/
project/scripts/daisyworld.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/docs/dai
syworld.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/docs/daisyworld.jl"
julia> █
```

Рисунок 4.8: Script:

Создадим скрипт `scripts/daisyworld-animate.jl` и загрузим его на фидеохостинг RUTUBE (рис. 4.9).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject$ cd scripts
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$ nano daisyworld-animate.jl
```

Рисунок 4.9: Создание скрипта

Создадим скрипт `scripts/daisyworld-count.jl` (рис. 4.10).

```
project/scripts$ nano daisyworld-count.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$ julia --project=. daisyworld-count.jl
Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/pr
ject/scripts$ █
```

Рисунок 4.10: Создание и выполнение:

Создание литературного стиля (рис. 4.11).

```
julia> Literate.markdown("daisyworld-count.jl", "docs")
^[[D^[[C^[[C[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.md"

julia> Literate.script("daisyworld-count.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count.jl"

julia> Literate.notebook("daisyworld-count.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count.jl`
[ Info: executing notebook `daisyworld-count.ipynb`
```

Рисунок 4.11: Литературный стиль:

Создание файла `scripts/daisyworld-luminosity.jl`, создание для него литературного стиля (рис. 4.12).

```
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx: ~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ nano daisyworld-luminosity.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx: ~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia
      Info: The latest version of Julia in the 'release' channel is 1.12.7+0.x64.linux.gnu.
      You currently have '1.12.5+0.x64.linux.gnu' installed. Run:

      juliaup update

in your terminal shell to install Julia 1.12.7+0.x64.linux.gnu and update the 'release' channel to that version.

      Documentation: https://docs.julialang.org
      Type "?" for help, "]" for Pkg help.
      Version 1.12.5 (2026-02-09)
      Official https://julialang.org release

julia> using Literate

julia> Literate.script("daisyworld-luminosity.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.jl"

julia> Literate.markdown("daisyworld-luminosity.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity.md"

julia> Literate.notebook("daisyworld-luminosity.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity.jl`
[ Info: executing notebook `daisyworld-luminosity.ipynb`
```

Рисунок 4.12: Создание файла и литературного стиля для него:

Выполнение скрипта `scripts/daisyworld-luminosity.jl` (рис. 4.13).

```

mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia --project=. daisyworld_luminosity.jl
Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$

```

Рисунок 4.13: Выполнение скрипта:

Создание и выполнение scripts/daisyworld__param.jl (рис. 4.14).

```

mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ nano daisyworld__param.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia --project=. daisyworld__param.jl
Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$

```

Рисунок 4.14: Создание и выполнение scripts/daisyworld__param.jl:

Создание литературного стиля scripts/daisyworld__param.jl (рис. 4.15).

```

object/scripts$ julia
Info: The latest version of Julia in the 'release' channel is 1.12.7+0.x64.linux.gnu.
You currently have '1.12.5+0.x64.linux.gnu' installed. Run:

juliaup update

in your terminal shell to install Julia 1.12.7+0.x64.linux.gnu and update the 'release' channel to that version.

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.
Version 1.12.5 (2026-02-09)
Official https://julialang.org release

julia> using Literate

julia> Literate.script("daisyworld__param.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld__param.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld__param.jl"

julia> Literate.markdown("daisyworld__param.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld__param.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld__param.md"

julia> Literate.notebook("daisyworld__param.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld__param.jl`
[ Info: executing notebook `daisyworld__param.ipynb`

```

Рисунок 4.15: Создани литературного стиля:

Создание и выполнение scripts/daisyworld-count__param.jl (рис. 4.16).

```

mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ nano daisyworld-count__param.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia --project=. daisyworld-count__param.jl
Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$

```

Рисунок 4.16: Создание и выполнение scripts/daisyworld-count__param.jl:

Создание литературного стиля scripts/daisyworld-count__param.jl (рис. 4.17).

```

julia> using Literate

julia> Literate.script("daisyworld-count__param.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.jl"

julia> Literate.markdown("daisyworld-count__param.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.md"

julia> Literate.notebook("daisyworld-count__param.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-count__param.jl`
[ Info: executing notebook `daisyworld-count__param.ipynb`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.ipynb`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-count__param.ipynb"

julia>

```

Рисунок 4.17: Создание литературного стиля scripts/daisyworld-count__param.jl:

Создание и выполнение scripts/daisyworld-luminosity__param.jl (рис. 4.18).

```

project/scripts$ nano daisyworld-luminosity__param.jl
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$ julia --project=. daisyworld-luminosity__param.jl
Activating project at `~/2026-1--study--simulation-modeling/labs/lab03/project`
mrshvecov@mrshvecov-HP-Laptop-15s-fq2xxx:~/2026-1--study--simulation-modeling/labs/lab03/project/scripts$

```

Рисунок 4.18: Выполнение и создание :

Создание литературного стиля scripts/daisyworld-luminosity__param.jl (рис. 4.19).

```

julia> using Literate

julia> Literate.notebook("daisyworld-luminosity__param.jl", "docs")
[ Info: generating notebook from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity__param.jl`
[ Info: executing notebook `daisyworld-luminosity__param.ipynb`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.ipynb`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.ipynb"

julia> Literate.script("daisyworld-luminosity__param.jl", "docs")
[ Info: generating plain script file from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.jl`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.jl"

julia> Literate.markdown("daisyworld-luminosity__param.jl", "docs")
[ Info: generating markdown page from `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/daisyworld-luminosity__param.jl`
[ Info: writing result to `~/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.md`
"/home/mrshvecov/2026-1--study--simulation-modeling/labs/lab03/project/scripts/docs/daisyworld-luminosity__param.md"

julia>

```

Рисунок 4.19: Создание литературного стиля scripts/daisyworld-luminosity__param.jl:

```
EditURL = "../scripts/daisyworld.jl"
```

““**example** daisyworld using DrWatson **quickactivate** «project» using Agents using DataFrames using Plots

```

include(sourcedir(«daisyworld.jl»))

using CairoMakie model = daisyworld()

daisycolor(a::Daisy) = a.breed

plotkwargs = ( agent_color=daisycolor, agent_size = 20, agent_marker = „☒“, heatarray =
:temperature, heatkwargs = (colorrange = (-20, 60),), ) plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5) plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)
step!(model, 40) plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)
save(plotsdir(«daisy_step001.png»), plt1) save(plotsdir(«daisy_step005.png»), plt2)
save(plotsdir(«daisy_step040.png»), plt3)

```

*This page was generated using [Literate.jl](https://github.com/fredr

```

````@meta
EditURL = "../daisyworld-count.jl"
```

````@example daisyworld-count
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(sourcedir("daisyworld.jl"))

using CairoMakie

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

model = daisyworld(; solar_luminosity = 1.0)

agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy count")
blackl = lines!(ax, agent_df[:, :time], agent_df[:, :count_black], col

```



```

whitel = lines!(ax, agent_df[!, :time], agent_df[!, :count_white], col
Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsiz
figure
@example daisyworld-count save(plotsdir("daisy_count.png"),
figure)

```

---

*This page was generated using Literate.jl.*

```
EditURL = "../daisyworld-count__param.jl"
```

““**example** daisyworld-count\_\_param using DrWatson **quickactivate** «project» using  
Agents using DataFrames using Plots using CairoMakie

```

include(sourcedir(«daisyworld.jl»))

black(a) = a.breed == :black white(a) = a.breed == :white adata = [(black, count), (white,
count)]

```

## 5 Параметры эксперимента

```
param_dict = Dict(:griddims => (30, 30), :max_age => [25, 40], :init_white => [0.2, 0.8],
:init_black => 0.2, :albedo_white => 0.75, :albedo_black => 0.25, :surface_albedo => 0.4,
:solar_change => 0.005, :solar_luminosity => 1.0, :scenario => :default, :seed => 165,)
```

## 6 Создаём список всех комбинаций

```
params_list = dict_list(param_dict)
for params in params_list

model = daisyworld(;params...)

agent_df, model_df = run!(model, 1000; adata)
figure = Figure(size = (600, 400));
ax = figure[1, 1] = Axis(figure, xlabel = "tick", ylabel = "daisy count")
blackl = lines!(ax, agent_df[:, :time], agent_df[:, :count_black], color = "black")
whitel = lines!(ax, agent_df[:, :time], agent_df[:, :count_white], color = "white")
Legend(figure[1, 2], [blackl, whitel], ["black", "white"], labelsize = 10)

plt_name = savename("daisy-count",params) * ".png"

save(plotsdir(plt_name), figure)

end
```

---

\*This page was generated using [Literate.jl](https://github.com/fredr)

```

```@meta
EditURL = "../daisyworld-luminosity.jl"
```

````@example daisyworld-luminosity
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots

include(sourcedir("daisyworld.jl"))

using CairoMakie

black(a) = a.breed == :black
white(a) = a.breed == :white
adata = [(black, count), (white, count)]

model = daisyworld(solar_luminosity = 1.0, scenario = :ramp)

temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]

agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)

```

```

figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
black1 = lines!(ax1, agent_df[:, :time], agent_df[:, :count_black], color = :black)
white1 = lines!(ax1, agent_df[:, :time], agent_df[:, :count_white], color = :white)
figure[1, 2] = Legend(figure, [black1, white1], ["black", "white"])

ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "luminosity")
lines!(ax2, model_df[:, :time], model_df[:, :temperature], color = :red)
lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity], color = :blue)
for ax in (ax1, ax2); ax.xticklabelsvisible = false; end
figure

save(plotsdir("daisy_luminosity.png"), figure)

```

This page was generated using Literate.jl.

```
EditURL = "../daisyworld-luminosity__param.jl"
```

```

““example daisyworld-luminosity__param using DrWatson quickactivate «project»
using Agents using DataFrames using Plots using CairoMakie
include(sourcedir(«daisyworld.jl»))
black(a) = a.breed == :black white(a) = a.breed == :white adata = [(black, count), (white,
count)]

```

7 Параметры эксперимента

```
param_dict = Dict( :griddims => (30, 30), :max_age => [25, 40], :init_white => [0.2, 0.8],  
:init_black => 0.2, :albedo_white => 0.75, :albedo_black => 0.25, :surface_albedo => 0.4,  
:solar_change => 0.005, :solar_luminosity => 1.0, :scenario => :ramp, :seed => 165, )
```

8 Создаём список всех комбинаций

```
params_list = dict_list(param_dict)
for params in params_list

model = daisyworld(;params...)

temperature(model) = StatsBase.mean(model.temperature)
mdata = [temperature, :solar_luminosity]

agent_df, model_df = run!(model, 1000; adata = adata, mdata = mdata)

figure = CairoMakie.Figure(size = (600, 600));
ax1 = figure[1, 1] = Axis(figure, ylabel = "daisy count")
blackl = lines!(ax1, agent_df[:, :time], agent_df[:, :count_black], color = :black)
whitel = lines!(ax1, agent_df[:, :time], agent_df[:, :count_white], color = :white)
figure[1, 2] = Legend(figure, [blackl, whitel], ["black", "white"])

ax2 = figure[2, 1] = Axis(figure, ylabel = "temperature")
ax3 = figure[3, 1] = Axis(figure, xlabel = "tick", ylabel = "luminosity")
lines!(ax2, model_df[:, :time], model_df[:, :temperature], color = :red)
lines!(ax3, model_df[:, :time], model_df[:, :solar_luminosity], color = :blue)
for ax in (ax1, ax2); ax.xticklabelsvisible = false; end
```

```
plt_name = savename("daisy-luminosity",params) * ".png"
save(plotsdir(plt_name), figure)

end
```

*This page was generated using [Literate.jl](https://github.com/fredr)

```
```@meta
```

```
EditURL = "../daisyworld__param.jl"
```

```
```
```

```
````@example daisyworld__param
```

```
using DrWatson
```

```
@quickactivate "project"
```

```
using Agents
```

```
using DataFrames
```

```
using Plots
```

```
using CairoMakie
```

```
include(sourcedir("daisyworld.jl"))
```

```
XXX
```

```
param_dict = Dict(
 :griddims => (30, 30),
```



```

:max_age => [25, 40],
:init_white => [0.2, 0.8],
:init_black => 0.2,
:albedo_white => 0.75,
:albedo_black => 0.25,
:surface_albedo => 0.4,
:solar_change => 0.005,
:solar_luminosity => 1.0,
:scenario => :default,
:seed => 165,
)

XXXXXXXX XXXXXXX XXXX XXXXXXXXXXXXXXXX

params_list = dict_list(param_dict)

for params in params_list

 model = daisyworld(;params...)

 daisycolor(a::Daisy) = a.breed

 plotkwargs = (
 agent_color=daisycolor,
 agent_size = 20,
 agent_marker = '█',
 heatarray = :temperature,
 heatkwargs = (colrange = (-20, 60),),
)

 plt1, _ = abmplot(model; plotkwargs...)

```

```

 step!(model, 5)
 plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs.

 step!(model, 40)
 plt3, _ = abmplot(model; heatarray = model.temperature, plotkwargs.

 plt1_name = savename("daisyworld",params) * "_step01" * ".png"
 plt2_name = savename("daisyworld",params) * "_step05" * ".png"
 plt3_name = savename("daisyworld",params) * "_step40" * ".png"

 save(plotsdir(plt1_name), plt1)
 save(plotsdir(plt2_name), plt2)
 save(plotsdir(plt3_name), plt3)

end

```

---

*This page was generated using Literate.jl.*

## 9 Выводы

В данной лабораторной работе мы попрактиковались в создании литературного  
стиля кода на примере Модели Daisyworld. # Список литературы{unnumbered}